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Clan culture and supply chain resilience in China

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ABSTRACT

This paper examines the effect of clan culture on supply chain resilience in China. We provide evidence that firms in cities with stronger clans experience fewer supply chain disruptions. The patterns remain robust to alternative measures, various subsamples, and instrumental variable estimations. When natural disasters, tariff shocks, and unfavorable credit policies negatively impact the stability of supply chains, our results show that clans significantly enhance resilience to these shocks. Mechanism tests suggest that clans serve as a substitute for contractual institutions and particularly, stabilize the relationships with socially connected suppliers.

1. Introduction

Firms organize production in a world of supply chains. The volume of trade in intermediate inputs is larger than trade in final goods and services (Minetti et al., 2019). Also, the division of design, production, distribution, and other phases along a supply chain can leverage the comparative advantage of firms. The fragmentation imposes significant search and customization costs on supply chain participants, as both customers and suppliers seek compatible partners to ensure business exchange. Despite the importance and the stickiness of supply chain relationships, disruptions have become the new normal in the past decade (Grossman et al., 2023).

A burgeoning literature examines a myriad of factors that promote firms to participate supply chains, the selection of suppliers, and their responsive arrangements under different exogenous shocks. For example, natural disasters, pandemic lockdowns, transportation disruption, unanticipated tariffs, and terrorist attacks can all disrupt supply chain relationships (Carvalho et al., 2021; Khanna et al., 2022; Pankratz and Schiller, 2024; Tan et al., 2024). The estimated losses from disruptions are approximately 42% of pre-tax earnings (Grossman et al., 2023). However, with few exceptions, there has been limited focus on how to enhance supply chain resilience, especially when firms encounter exogenous shocks.

A supply chain relationship is not only governed by a contract but also maintained by social capital. Incomplete contract leads to frequent renegotiation and hinder deep cooperation, but social connections can act as an effective mechanism to enhance mutual trust and mitigate opportunism, and thereby promote the establishment of supply chain relationships (Chen et al., 2021; Dasgupta et al., 2021). Also, socially embedded supply chains—a mode of "doing business with friends"— show great resilience under different conditions (Ding et al., 2023).

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Notably, China is a family- and relationship-centered society (Fan et al., 2023). For example, hometown favoritism is prevalent in various areas, including fiscal expenditure, bank loan resources, and corporate investments (See, e.g., Fisman et al., 2018; Faccio and Parsley, 2009; Persson and Zhuravskaya, 2016; Guo et al., 2021). This favoritism arises from traditional values and norms in China, where people feel a moral obligation to support and reward their hometowns or fellow townsmen. In discussing Chinese traditional culture, clan culture is indispensable. Clans are an incredibly important social organization built on kinship, consisting of agnate households with one common ancestor (Freedman, 1966; Greif and Tabellini, 2017). Clan identity is inherently determined for everyone, which fosters deep-rooted familial and communal bonds among members. By its nature, clan connections are even stronger and more stable than hometown ties.

Clans, as grassroot organizations in pre-modern society, often substitute government functions in several areas. They were responsible for collecting taxes, providing public services such as building road and schools, and settling disputes within the lineage group or conflicts with outsiders (Chen et al., 2021; Greif and Tabellini, 2017). These functions have persisted into modern society, for example, strong clans can increase the participation of villagers in voting and the expenditure on public goods (Su et al., 2011; Xu and Yao, 2015).

In this paper, we provide evidence that strong clan strength can enhance the supply chain resilience of local firms. Participation in supply chains leaves firms exposed to risks across multiple dimensions. Shocks to suppliers like extreme weather or terrorist attacks, inevitably and negatively impacting their performance, are far away from the place of consumption. The lack of timely support for partners in difficult episodes may lead to relationship breakup. Also, contractual incompleteness, associated with weak institutional environment in China, is likely to encourage opportunistic behaviors and discourage durable relationships.

We posit that the norms and social capital embedded in traditional clans can help address these issues and reduce the possibility of relationship disruptions. First, both institutional elements and moral obligations rooted in Confucian clans strongly sustain cooperation. Clans typically establish their own rules to organize the group and extend punishments for certain behaviors of their members. Additionally, harming outsiders is regarded as damaging the clan's reputation (Liu, 1959), and opportunistic behaviors like cheating and free riding are considered as morally unacceptable in traditional values (Greif and Tabellini, 2010). Thus, supply chain relationships with entities in regions with strong clan strength are less likely to face risks of contract non-compliance. Second, institutionalized clans provide informal arrangements to facilitate risk-sharing (Chen et al., 2024b). In pre-modern society, clans served as a safety net by doling out financial and physical provision during famines and taking care of aged and disabled members in daily life (Cao et al., 2022). When exogenous shocks negatively affect suppliers' performance, firms from regions with intensive clan culture may have a heightened commitment to overcome difficulties with their partners together. Moreover, strong clans have the capacity to provide financial or social resources to support connected firms if shocks propagate along the supply chain and adversely affect the operations of customers. These characteristics are expected to stabilize supply chain relationships when firms face high risks.

However, on the downside, several characteristics of clans can increase the vulnerability of supply chains. The intra-group trust, rather than generalized trust, fostered by clan culture may lead firms to favor partners within their social network. This preference can constrain supplier diversity and potentially lower partner selection standards, thereby undermining effective supply chain management. Also, the hierarchical and rigid norms embedded in clans inherently suppress risk-taking activities, including corporate innovation (Huang et al., 2022). This may hinder firms' long-long sustainable growth, and weakens their ability to withstand exogenous shocks.

To test these hypotheses, we carry out the empirical analyses along two lines. We begin by examining the impact of clan culture on supply chain disruptions of local firms. To clearly identify the stabilization effect, we then exploit various shocks that disrupt supply chains and examine whether clans enhance resilience. We measure the strength of clan culture using the number of genealogies stored in each prefecture, defined as the logarithm of the number of genealogies compiled before 1990 normalized by the 1990 population. A genealogy is a book that chronicles the history of a clan, including their lineages from the earliest common ancestors, details of marriages and offspring, notable accomplishments of members, and standards of conduct. Compiling genealogies is an important dimension of clan activities. To measure supply chain resilience, we utilize the firm-to-firm supply chain data from the FactSet Revere over the period of 2007-2021, and primarily focus on separation rates, defined as the number of terminated suppliers divided by the average number of total suppliers in the current and preceding years.

Our analyses show that firms operating in cities with denser genealogies experience significantly lower separation rates in their supply chains. The economic magnitude is nonnegligible: a one-standard-deviation increase in the density of genealogies will reduce the separate rate by 13.96% on average. Using the stratified COX proportional hazards model, we further show that the strength of local clans lengthens the duration of each supply chain relationship. To ensure our findings, we turn to a supplier-customer-year dataset and regress the probability of disruption for each relationship on clan density. After controlling for firm, supplier-year, province-year, and industry-year fixed effects and a variety of firm and city correlates, we consistently find that clan density significantly decreases the likelihood of supply chain dissolution.

Additionally, our findings remain robust when we adopt alternative measures of supply chain resilience, assess the clan density using the genealogies compiled before 1912 and 1949, cluster standard errors at different levels, and utilize different subsamples that might be subject to identification issues. To formally address the endogeneity concerns, we instrument the strength of clans in each prefecture using rice-wheat ratio, agricultural potential yield, and the number of chaste women. IV estimations provide consistent evidence that higher clan density reduces separation rates.

Next, we examine how clans stabilize supply chain relationships under different types of exogenous shocks. We first pick one of the most significant natural disasters—earthquake, which has been proved to cause substantial supply chain disturbance (Inoue and Todo, 2019; Carvalho et al., 2021). We first confirm a positive association between earthquake occurrences and supply chain disruptions of local firms, and then show that high clan density significantly mitigates the adverse effect of unexpected natural disaster shocks.

Credit shocks represent another source of disturbance. Partners always provide liquidity to each other, which allows firms to ease credit constraints by participating in various phases of supply chains (Minetti et al., 2019). Correspondingly, when firms face liquidity shortage, partners are likely to terminate relationships for insufficient financing resources. As *the Guideline on Green Credit* issued by China Banking Regulatory Commission (CBRC) restricts bank loan extension to "two high, one overcapacity" industries, we exploit the introduction of the Guideline as a natural experiment and classify firms in the targeted industries as the treatment group and others as the control. We find that guideline-credit-constrained firms have significantly higher separation rates in their supply chains following its implementation. Particularly, strong local clans enhance supply chain resilience to the credit policy shock.

We then focus on the tariff shock arising from the international market. During the China-U.S. trade war starting in 2018, the Chinese government implemented retaliation tariffs, which increased the average tariff on the U.S. goods from 7.8% in 2017 to 25.9% in 2019. The sharp increase in input tariffs affect production costs and prompt firms to renegotiate with existing suppliers or seek substitutes. Our analysis shows that firms having relationships with U.S. suppliers whose products are targeted by retaliation tariffs, experience significantly higher foreign separation rates, while they sustain their production by reducing domestic separation rates. Clans provide greater resilience value to domestic supply chains in the face of tariff shocks.

Finally, we explore two mechanisms through which traditional clans create resilience value. We show that clans play a more important role in stabilizing supply chains in regions with weak institutional environment. We interpret these findings as evidence that institutional and moral elements of clans create favorable business environment and enhance the attractiveness of local firms as trading partners. We also discuss the role of social connections in stabilizing supply chains. We classify relationships into connected and non-connected ones based on whether the CEO or chairpersons of customers and suppliers share hometown ties and whether they share the same surnames. During both ordinary and challenging periods, clans primarily enhance the resilience of socially connected supply chains.

Our work deepens the understanding of the persistent impact of historical institutions, and more broadly, how cultural factors affect socioeconomic outcomes (e.g., Chen et al., 2020; Dell, 2010; Greif and Tabellini, 2017; Guiso et al., 2006, 2016; Tabellini, 2008). Clans have been a research focus in the context of China. As grassroot kinship-based organizations, clans offer informal channels for financing and information resources, which promotes the development of private sector but hinders the growth of modern banks (Chen et al., 2021; Zhang, 2020); clans also contribute to public governance and facilitate risk-sharing, for example, saving lives and reducing violence in famines (Cao et al, 2022; Chen et al., 2024a; Xu and Yao, 2015). Our work shifts the focus from collectivism and economic performance at the region level to corporate behaviors. Leveraging the firm-to-firm supply chain data, we provide evidence that clans can enhance the stability of business relationships. This also offers new insight into responsive policies for stabilizing macroeconomy in the episodes of heightened geopolitical conflicts.

Our work also contributes to the emerging literature on supply chains by highlighting the resilience value created by traditional culture and norms. A few studies model the formation and equilibrium of supplier network and investigate the motives of firms to participate in supply chains (Elliott et al., 2022; Minetti et al., 2019), and other studies have identified various determinants of supply chain disruptions, including climate change, economic and political uncertainties, transportation failure, anticipated tariffs (Boehm et al., 2019; Balboni et al., 2023; Charoenwong et al., 2023; Dong et al., 2022; Grossman et al., 2024; Pankratz and Schiller, 2024), and among others. There is still limited evidence on how to enhance supply chain resilience. One exception is Grossman et al. (2023) who evaluate the efficacy of government policies in face of global supply chain disruptions. Khanna et al. (2022) also study firm characteristics that increase resilience, such as product differentiation. We provide a new perspective of cultural trait to help firms stabilize supply chain relationships. Additionally, by examining the resilience value of clans in the face of natural disasters, tariff increases, and credit shocks, our work relates to the research on the propagation of shocks along the supply chain and aligns with studies on the impact of China-U.S. trade war (Alfaro et al., 2021; Fajgelbaum and Khandelwal, 2022).

The remainder of this paper is organized as follows. Section 2 introduces the background. Section 3 describes data, variables, and empirical strategies. In Section 4, we estimate the impact of clan density on supply chain disruptions. In Section 5, we provide evidence that clans enhance supply chains resilience to three types of exogenous shocks. Section 6 turns to mechanism tests. Section 7 provides further discussion on supply chain stability. Section 8 gives concluding remarks.

2. Background

2.1. Clans in ancient China

Cultural factors capable of forming social organizations are of paramount importance, as they might consist of the most deep-rooted and stable elements of the society. Among others, the clan is an informal organization comprising a network along with individuals linked by kin-based bond with actual blood ties and fictive kinship (Collins, 2004). While there is evidence that clans are embedded in governance structures in regions such as Central Asia and Saudi Arabia (Kraft, 2003; Hudson et al., 2015), the significance of clan in China is far greater than any other countries.

The clan in traditional China does not completely resemble in other societies, as it embodies both the characteristics of the clan as it exists elsewhere and the distinct nature of the Chinese joint family (Liu, 1959). Established early in Zhou Dynasty since 1122 B.C., the clan system was initially created to secure the prestigious social status of the nobility. It was abolished during the Tang Dynasty but later revived in a new form proclaimed by the Song Neo-Confucian theories again. Since then, the clan was no longer exclusively associated with privileged class; instead, it came to include families from various social strata. In this way, the clan becomes the broadest organizational basis in ancient China, continuously and dynamically shaping social orders, institutional structures, and moral systems.

In addition to being a unilinear descent group, clans in China also fulfill many functions related to education, ceremony, social security, the maintenance of law and order, and the preservation of property rights (Fei and Liu, 1982). This complex of function is sustained through regular activities, such as compiling genealogies, building ancestral halls, holding ritual gathering, establishing clan rules, and enforcing ritual penalties. Particularly, clans functioned as a complementary institution that assisted imperial power. In ancient China, the southern part, especially those along the coastal areas, are far away from the central authority, which encouraged the development of unified societal organizational forms. This provides the rationale to the geographic variation of clans: they were rather weak in northern China, moderately developed in central China, and very strong in southern China (Liu, 1959).

2.2. Clans and business in modern China

Like other traditional Chinese institutions, clan experienced waves of shocks caused by the radical changes over the past two centuries in China. Irrevocably decayed as it could be, the clan never disappeared in modern China. Even though some of its functions have been largely replaced by modern government, the value systems and beliefs repeatedly enhanced by the clan institution persist. Several characteristics of the modern clan can potentially generate both positive and negative impacts on firm sustainability and supply chain dynamics in our context.

In terms of potential benefits, group cohesion and information sharing embedded in clan organizations are crucial for business stability. Blood relationships on the paternal side are believed to form a natural affinity among clan members. Ancestral rites then reminded clan members that, as descendants of a common ancestor, they should regard one another as if they were still members of the same family. Within the familial context, trust is inherently stronger than among unrelated individuals. In business, trust translates into lower transaction costs, reduces risk of contractual breach, and sustains long-term cooperation. For example, entrepreneurs from Fujian and Guangdong provinces, the representative regions of strong clans, are highly united and exert considerable influence when doing business overseas. Typical clan activities, such as compiling genealogy compilation or engaging in ancestral worship, provide valuable opportunities for information sharing. This is crucial to establish business relationship both within the group and with others recommended by clan members.

Clan rules, established to organize the group and monitor the behavior of clan members, prevents members from harming outsiders in a way that could damage the clan's reputation (Liu, 1959). Hence, people raised with clan culture are less likely to engage in opportunistic behaviors that negatively affect business relationships. Also, the collectivism associated with group cohesion prevent outsiders from harming the interests of clan members, as such actions are perceived as detrimental to the group. This further stabilizes the relationship with entities from regions with strong clan influence. Moreover, clans can serve as a mechanism for risk resistance. In pre-modern society, clans served as a safety net in difficult times by doling out financial and physical provision during famines and taking care of aged and disabled members (Cao et al., 2022). Socialized by such culture framework, firms in areas with deep-rooted clan traditions are more likely to stand by their partners and engage in cooperative problem-solving under exogenous shocks.

On the downside, several characteristics of clans may increase relationship vulnerabilities. First, clan culture fosters societal trust, but primarily in the form of intra-group trust rather than generalized trust (Fukuyama, 1996; Chen et al., 2022; Fan et al., 2025). When firms from regions with strong clans attempt to establish relationships, they are more likely to seek partners within their social network. This preference entails two possible challenges. The first is limited supply chain diversification. Prior studies have shown that supply chains are more resilient to exogeneous shocks when firms have a larger supplier network size, as it is less costly to substitute across suppliers (Elliott et al., 2022; Khanna et al., 2022). Therefore, limiting supplier sources to close-knit social networks may be detrimental to effective risk management. The second challenge is that close relationship may blur the criterial for supplier selection. Deep embeddedness in an organization leads firms to lower standards and favor intra-group partners, thereby increasing the vulnerability of relationships to exogenous shocks.

Second, clans established hierarchical and rigid rules and norms (Landes, 2006; Zhang, 2020), which by their nature tend to suppress innovation and discourage risk-taking activities. Individuals deeply influenced by clan culture often exhibit weak contrarian thinking or a reluctance to challenge established rules. Directors or managers from regions with strong clan culture are more conservative and less likely to undertake high-risk investments like corporate innovation or cross-region mergers (Huang et al., 2022). Although firms may maintain relatively stable business network in the short run, the absence of ongoing innovation will hinder long-term development and gradually erode their market competitiveness. Then, firms themselves have limited capacity to withstand external shocks, and partners may sever supply chain relationships with firms that exhibit unfavorable competitive edge.

3. Data and empirical design

3.1. Sample and key variables

The main dataset for our empirical analysis is constructed by combining several data sources: supply chain data from the FactSet Revere, genealogy data from *The General Catalog of Chinese Genealogies* compiled by the Shanghai library, financial statement information, corporate governance, as well as manager and director information of listed firms from the China Stock Market and Accounting Research (CSMAR) database. By excluding listed firms without any records in the FactSet, we obtain a final sample consisting of 2,833 Chinese listed firms over the period of 2007-2021.

Supply chain resilience. We collect the firm-to-firm supply chain data from the FactSet Revere. This database covers 90,292 suppliers and 147,068 customers from 177 countries during our sample period and has been extensively used in recent works on supply chains (Charoenwong et al., 2023; Ding et al., 2023; Pankratz and Schiller, 2024). It provides specific start and end dates for each relationship, which is important to our analysis of supply chain resilience under external shocks. We filter the database by selecting Chinese listed firms based on ISIN identifier that are marked as customers. In total, we obtain 2,833 firms establishing relationships with 2,322 unique domestic suppliers, comprising 32,375 customer-supplier-year observations over the period of 2007-2021.

Our primary measure of supply chain resilience is the separation rate defined as follows:

$$Separation\ Rate_{it} = \frac{N\ of\ suppliers\ to\ i\ in\ year\ t-1,\ who\ don't\ supply\ in\ t}{\{(N\ of\ suppliers\ to\ i\ in\ year\ t-1) + (N\ of\ suppliers\ to\ i\ in\ year\ t)\}/2}, \tag{1}$$

where i and t denote firm and time, respectively. This measure reflects the number of suppliers terminated by firm i relative to the average number of total suppliers in year t-1 and year t.

To further capture the resilience of each supply chain relationship, we construct two other variables using the customer-supplier cross-sectional dataset and customer-supplier-year panel. One is the duration of relationship ($Duration_{ij}$), proxied by the number of years that customer i and supplier j maintain their relationship. The other is a disruption dummy ($Disrupt_{ijt}$), indicating whether the supply chain relationship between firms i and j is terminated at year t. In the robustness, we also adopt the log transformation and the inverse hyperbolic sine (HIS) transformation to the number of terminated suppliers, and calculate the separate rate by replacing the denominator in Eq. (1) with the number of suppliers in the current year.

Clan culture. We measure the strength of clans using the number of genealogies. A genealogy is a book that chronicles the history of a clan, including their lineages from the earliest common ancestors, details of marriages and offspring, notable accomplishments of members, infamous ex-members, properties, and standards of conduct. The compilation of genealogy is considered an essential component of clan culture which plays a pivotal role in regulating human behavior, bolstering belonging, and promoting cooperative spirit within a group based on kinship (Chen et al., 2024a). The number of genealogies reflects the degree of activity and the local influence for each clan, and the measure has been commonly used in prior studies on clans (e.g., Bisin and Verdier, 2024; Fan et al., 2023).

We manually collected the genealogy data from *The General Catalog of Chinese Genealogies*, compiled by the Shanghai Library and published by the Shanghai Ancient Books Publishing House in 2008. For each genealogy, we extract the surname, compilation year, and location information. After excluding genealogies with incomplete information (missing location and/or compilation year) and those stored outside the mainland China, we obtain a set of 43,382 genealogies across 278 prefectures, of which 36,102 are compiled before year 1990. To mitigate the concern that the level of regional economic development might affect both the compilation willingness and firms' business relationship, we employ the logarithm of the number of genealogies compiled before 1990, a period predating China's economic boom, to proxy for clan density in each prefecture.

3.2. Empirical specifications

To investigate the impact of clan culture on supply chain stability, we first estimate the specification using a firm-year panel as follows:

$$Separation\ rate_{it} = \alpha + \beta Clan_{ic,t-1} + Controls + \gamma_{jt} + \delta_{pt} + \varepsilon_{it}, \tag{2}$$

where i, j, c, p, and t correspond to firm, industry, prefecture, province, and time, respectively. The dependent variable is the separation rate of firm i in prefecture c at year t, defined in Eq. (1). $Clan_{ic,t-1}$ is the logarithm of the cumulative genealogy number per 10,000 persons in firm i's locality, prefecture c, at year $t-1$ [1]. A negative coefficient β implies that strong clans can hinder supply chain disruptions.

We include a set of control variables that may affect relationship survival, including firm's age, employees, leverage, ROE, cash ratio, fixed asset ratio, the shareholding by the largest shareholder, a dummy for state-controlled firms, a dummy for foreign ownership, and CEO duality. To account for regional variations of economic development, we include prefecture-level GDP per capita, fiscal expenditure, education expenditure, the ratio of bank credit to GDP, and the number of industrial firms. Geography and climate conditions which affect regional factor endowment and comparative advantage are also correlated with both clan strength and economic activities (Chen et al., 2022). Thus, we further control for land size, distance to the coast, average slope, and river density of firms' locality. All independent variables are with one-year lag. Industry-year and province-year fixed effects, γ_{jt} and δ_{pt}, are included in all regressions. Standard errors are clustered at the firm level.

To further capture the resilience value of clan culture, it is essential to investigate the changes of supply chains in response to exogenous shocks across firms from regions with varying levels of clan density. Then, we augment Eq. (2) and estimate the following specification:

$$\begin{aligned} Separation\ rate_{it} = {} & \alpha + \beta_1 Shock_{it} + \beta_2 Clan_{ic,t-1} \times Shock_{it} \\ & + Controls + \gamma_j + \delta_c + \tau_t + \varepsilon_{it}, \end{aligned} \tag{3}$$

$Shock_{it}$ in Eq. (3) represents three types of shocks that affect the stability of supply chains, earthquake, credit, and tariff shocks. We employ a difference-to-differences (DID) framework to define the variables of shock.

[1] Due to the relocation of firms, the strength of clans for each firm's location is a time-varying variable.

Table 1
Summary statistics.

Variables	Observation	Mean	SD	Median	Min	Max
Dependent Variables						
Separation Rate	26,400	0.050	0.193	0	0	2
Duration	6,697	1.956	1.509	1	1	14
Disrupt	32,375	0.150	0.357	0	0	1
Independent Variables						
Clan	26,400	-1.485	1.662	-1.407	-6.019	3.630
Number of Genealogies	26,400	296.605	485.998	68	0	3230
Firm Controls						
Age	26,400	2.807	0.38	2.833	0	3.738
Foreign	26,400	0.049	0.216	0	0	1
SOE	26,400	0.437	0.496	0	0	1
Dual	26,400	0.245	0.430	0	0	1
Leverage	26,400	0.488	5.400	0.455	0	877.256
ROE	26,400	0.045	1.329	0.076	-174.900	33.831
Profit	26,400	0.271	0.237	0.239	-24.501	1.154
Size	26,400	7.637	1.288	7.626	1.946	12.785
Cash	26,400	0.164	0.147	0.126	-0.165	10.309
Fixed Asset	26,400	0.229	0.173	0.193	0	0.971
First Holder	26,400	35.093	15.293	32.980	0.290	100
City-level Geography Controls						
Land Size	26,400	9.115	0.769	9.196	7.110	12.030
ln Distance to Coast	26,400	4.769	1.446	5.100	0.515	7.921
River Density	26,400	0.422	0.354	0.308	0.022	1.399
Slope	26,400	8.910	5.137	9.144	1.592	22.998
City-level Economic Controls						
ln GDPpc	26,400	14.346	11.442	11.386	0.415	53.235
Loan/GDP	26,400	2.089	1.272	1.835	0.0373	8.871
ln General Budget Expenditure	26,400	16.140	1.259	16.058	11.271	18.25
ln Education Expenditure	26,400	14.314	1.187	14.214	9.323	16.256
ln Firm Number	26,400	7.977	1.045	8.086	3.091	9.841
Instrument variables						
Arable land potential productivity	26,400	7.775	0.936	7.949	0	9.191
Chaste Women	26,400	1.560	1.176	1.792	0	3.738
Rice Wheat Ratio	26,384	9.738	6.506	9.210	-4.374	22.050

Notes: This table presents descriptive statistics of main variables. Our sample consists of 2,833 listed firms over the period of 2007-2021. The variable definitions can be found in Appendix A1.

The first is the earthquake shock. For each earthquake, we classify firms into two groups: i) the treatment group, comprising firms whose location is affected by the earthquake; ii) the control group, comprising firms that did not experience any earthquakes throughout the entire sample period. We define the estimation window as [-5, 5], with year 0 representing the earthquake event year. These procedures allow us to construct a new dataset with rolling estimation windows based on the timing of each earthquake. Then, the variable $Earthquake_{it}$ is a DID interaction term between a treatment variable, indicating whether firm i's location is affected by an earthquake, and a dummy variable identifying the five-year post-earthquake period.

The second type is the credit policy shock. Access to bank credit is an important determinant of firms' choices whether to participate supply chains (Minetti et al., 2019). We focus on a special credit policy issued by China Banking Regulatory Commission, *the Guideline on Green Credit*, which restricts bank lending to firms in heavily polluting, resource-intensive, and over-capacity sectors. We again define an estimation window of [-5, 5], with time 0 corresponding to the policy implementation year, 2012. In this context, the variable $Shock_{it}$ in Eq. (3) refers to $GCguide_{it}$, an interaction term between a treatment dummy, which equals one if firms belong to credit-constrained industries targeted by the *Guideline* and zero otherwise, and a post dummy for post-policy period.

The third type is the tariff shock. In this part of analysis, we exploit 2018-19 China-U.S. trade war, the most significant trade conflict in this century. Similarly, we classify firms into a treatment group—those with at least one U.S. supplier whose products are subject to the retaliatory tariff increases imposed by the Chinese government—and a control group comprising all other firms. Accordingly, $Tradeshock_{it}$ is constructed as an interaction term between a dummy variable indicating that firms' sourcing is negatively affected by the trade war and a post-treatment dummy that equals one for years after the tariff escalation and zero for the five preceding years (2013-2017).

In the following analysis, we will first test whether these shocks negatively affect supply chain disruptions in turn. The variable of our particular interest in Eq. (3) should be the interaction of $Clan_{ic,t-1} \times Shock_{it}$. A negative sign of β_2 is consistent with our expectation that clan density can mitigate the adverse impact of exogenous shocks on supply chain stability. The control variables are the same as those in Eq. (2).

3.3. Summary statistics

Our final sample consists of 26,400 firm-year observations for the period of 2007-2021. Table 1 reports the summary statistics for

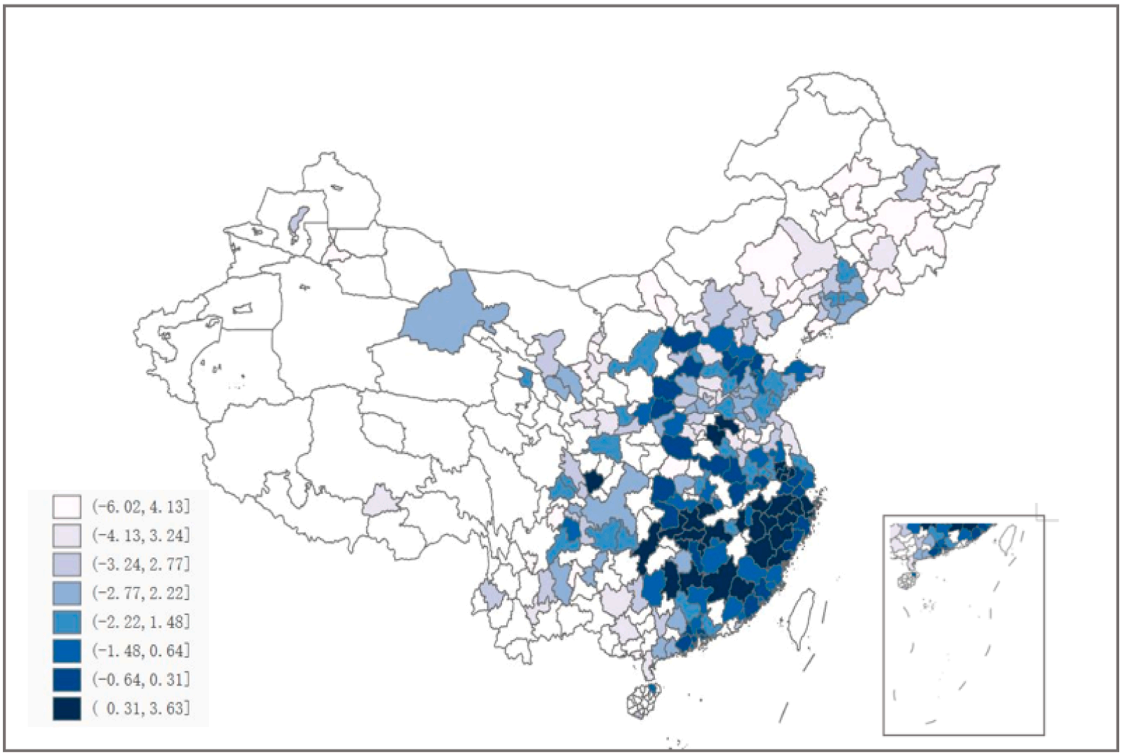

Fig. 1. Geographical distribution of clans.
Notes: The strength of clans is measured by the logarithm of the number of genealogies per 10,000 persons in each prefecture. A prefecture with more (fewer) genealogies is shaded in darker (lighter) blue. Regions without available data on the 1990 population or without genealogies compiled before 1990 are colored in white.

main variables. We observe significant variations of clan density across regions. The average number of genealogies is 296.6, while the maximum is 3,230 and the minimum is 0. In our dataset, genealogies complied before 1990 could be found in 218 prefectures in mainland China, among which 56 prefectures have more than 100 editions. Fig. 1 depicts the geographical distribution of genealogy density, suggesting that clans are concentrated in southern and southeastern China although some cities in the northeast still maintain a high number of genealogies. The variations provide a good context to examine firms' behaviors across subnational regions. We exclude coastal provinces in the robustness check to address the concern of potential sample bias.

As for supply chain stability, the statistics show that Chinese firms experience an average separation rate of 5%. The mean duration for all supply chain relationships in our sample is 1.956 years, indicating a lack of "stickiness" mentioned in Grossman et al. (2024) and highlighting a high level of vulnerability. Consequently, to identify factors affecting supply chain stability and particularly, those enhancing supply chain resilience provide important managerial implications for Chinese firms.

4. Baseline results: clan culture and supply chain disruptions

4.1. Benchmark

Column 1 of Table 2 reports the estimation results of Eq. (2). Drawing on the firm-year dataset, we show that the strength of clan culture significantly decreases supplier separations of local firms after controlling for a variety of firm and city controls, as well as industry-year and province-year fixed effects. The estimated coefficient before *Clan* is negative and statistically significant at the 1% level, suggesting that clans, as expected, can help firms stabilize their supply chains. The effect is economically sizable: a one-standard-deviation increase in genealogy density at the firm's locality decrease the separate rate by 13.96% (=0.0042 × 1.662/0.05) on average.

To further capture the stabilization effect of clan culture, we focus on the duration of each supply chain relationship. Using the supplier-customer pair-level cross-section dataset and employing a stratified COX proportional hazards model, we show that strong clan strength significantly lengthens the relationship duration. As shown in column 2, a one-unit increase in clan density is significantly associated with a 0.05-year increase in the relationship duration.

Moreover, we examine the role of clan culture using a highly disaggregated supplier-customer-year dataset and take the dummy indicating relationship disruption as the dependent variable in a linear probability model (Ding et al., 2023). As the disruption might arise either from the customer or the supplier decision, we can control for a supplier-year fixed effect in this dataset to better isolate the

Table 2
Baseline results: clan density and supplier disruptions.

	Model 1: Firm-Year Fixed Effect Model	Model 2: Cox Proportional Hazards Model	Model 3: Customer-Supplier-Year Fixed Effect Model	
	Separation Rate	Duration	Disruption dummy	
	(1)	(2)	(3)	(4)
Clan	-0.0042***	0.0511**	-0.0381***	-0.0236*
	(0.002)	(0.024)	(0.011)	(0.014)
Age	0.0090**	0.1761**	-0.0757	0.0608
	(0.004)	(0.087)	(0.056)	(0.099)
Foreign	-0.0122**	0.0571	0.0059	0.0144
	(0.005)	(0.111)	(0.035)	(0.050)
SOE	0.0145***	0.0559	0.0187	0.0294
	(0.003)	(0.060)	(0.018)	(0.024)
Dual	-0.0031	-0.0662	0.0107	0.0038
	(0.003)	(0.057)	(0.010)	(0.013)
Leverage	0.0001**	0.1730	0.0030	0.0299
	(0.000)	(0.172)	(0.030)	(0.045)
ROE	-0.0009*	0.0128	-0.0023	-0.0009
	(0.000)	(0.114)	(0.002)	(0.002)
Profit	-0.0007	-0.1990	0.0211	-0.0412
	(0.007)	(0.239)	(0.029)	(0.043)
Size	0.0098***	-0.0994***	0.0126**	0.0031
	(0.001)	(0.027)	(0.006)	(0.010)
Cash	0.0112	-0.1599	-0.0036	-0.0517
	(0.008)	(0.254)	(0.040)	(0.059)
Fixed Asset	-0.0119	0.0119	-0.0264	-0.0318
	(0.009)	(0.252)	(0.039)	(0.058)
First Holder	-0.0001	0.0005	0.0001	0.0006
	(0.000)	(0.001)	(0.000)	(0.001)
ln GDPpc	-0.0001	-0.0000	0.0000	0.0000
	(0.000)	(0.000)	(0.000)	(0.000)
Loan/GDP	-0.0083*	0.0026	-0.0040	-0.0022
	(0.005)	(0.042)	(0.012)	(0.016)
ln General Budget Expenditure	0.0082	-0.0864	-0.0375	-0.0430
	(0.010)	(0.142)	(0.045)	(0.060)
ln Education Expenditure	0.0008	0.0523	0.0060	-0.0151
	(0.011)	(0.151)	(0.043)	(0.057)
ln Firm Number	0.0025	-0.0642	0.0268	0.0528
	(0.005)	(0.059)	(0.028)	(0.040)
Land Size	-0.0037	-0.0163	-0.0646	-0.1325**
	(0.005)	(0.066)	(0.042)	(0.052)
ln Distance to Coast	0.0007	0.0196	-0.0389	0.0461
	(0.002)	(0.021)	(0.030)	(0.043)
River Density	-0.0135	-0.1992*	-0.1716	-0.1028
	(0.015)	(0.105)	(0.210)	(0.314)
Slope	0.0009	-0.0096	-0.0031	0.0022
	(0.001)	(0.007)	(0.007)	(0.009)
Industry-Year FE	Yes	-	Yes	Yes
Province-Year FE	Yes	-	Yes	Yes
Firm FE	-	-	Yes	Yes
Supplier-Year FE	-	-	-	Yes
Customer Industry Strata	-	Yes	-	-
Supplier Strata	-	Yes	-	-
Year Strata	-	Yes	-	-
Observations	26,299	6,071	31,556	23,004
Adj R^2	0.041		0.053	0.252

Notes: This table reports the regressions on the effect of clan density on supply chain stability of local firms. In column1, we take separate rates defined in Eq. (1) as the dependent variable and estimate Eq. (2) using a firm-year panel. In column 2, we adopt the Cox Proportional Hazards Model and estimate the effect of clan density on relationship duration using a cross-sectional dataset. In columns 3-4, we use the customer-supplier-year dataset and take a dummy indicating the relationship disruption as the dependent variable. Columns 1 control for province-year, and industry-year fixed effects; column 3 adds a firm fixed effect; column 4 additionally control for supplier-year fixed effects. *Clan* is calculated as the logarithm of one plus the cumulative number of genealogies before 1990 normalized by the 1990 population for each prefecture. The definitions of other variables can be found in Appendix Table A1. Robust standard errors clustered at the firm level in columns 1-2 and clustered at the customer-supplier pair level in columns 3-4 are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

effect of clans. The results in columns 3 and 4 reinforce our baseline findings. Firms from cities with stronger clans experience significantly fewer supplier separations.

As for the controls, foreign ownership provides certain comparative advantage and maintain stable supply chains, while large-sized

Table 3
Robustness: alternative measures.

Panel A: Supply chain disruption				
	Log number of terminated suppliers (1)	HIS of terminated domestic suppliers (2)	Dummy (3)	Separation rate (normalized by existing suppliers) (4)
Clan	-0.0082***	-0.0105***	-0.0082***	-0.0065***
	(0.003)	(0.004)	(0.003)	(0.002)
Controls	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
Observations	26,299	26,299	26,299	26,299
Adj R^2	0.116	0.116	0.111	0.056
Panel B: Clan strength				
	(1)	(2)	(3)	
Clan all	-0.0036**			
	(0.002)			
Clan before 1912		-0.0041***		
		(0.002)		
Clan before 1949			-0.0040***	
			(0.002)	
Controls	Yes	Yes	Yes	
Industry-Year FE	Yes	Yes	Yes	
Province-Year FE	Yes	Yes	Yes	
Observations	26,299	26,299	26,299	
Adj R^2	0.041	0.041	0.041	

Notes: In Panel A, we employ different measures of supply chain disruptions. The dependent variable in columns 1-2 are the natural logarithm form and HIS form of terminated suppliers; the dependent variable in column 3 is an indicator of whether the firm has terminated at least one supplier; the dependent variable in column 4 is the separation rate calculated by dividing the number of terminated suppliers by the total number of existing suppliers. In Panel B, we employ different measures of clan strength. *Clan all* is the logarithm of one plus the number of all the genealogies normalized by the 1990 population for each prefecture; *Clan before 1912* and *Clan before 1949* are the logarithm of one plus the cumulated number of genealogies written before 1912 and 1949 normalized the 1990 population, respectively. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

firms show higher separation rates and shorter length of relationship. Several studies also examine supply chain resilience along dimensions other than separations, for example, Khanna et al. (2022) explore entry rate, that is, the establishment of new relationships. In Appendix Table A2, we measure the supply chain resilience using the number of new suppliers relative to the average number of total suppliers in the current and preceding years. The results show that firms in regions with strong clans are less likely to establish relationships with new suppliers. Overall, clans help firms retain existing suppliers to ensure production continuity.

4.2. Robustness checks

We perform the first set of robustness checks using alternative measures of supplier disruptions. Instead of separation rate, we focus on the number of terminated suppliers by applying the log transformation and the inverse hyperbolic sine (HIS) transformation in columns 1 and 2 of Panel A, Table 3, respectively. We consistently find that clan density significantly decreases supplier separations. In column 3, we construct a dummy, which equals one if the firms terminate at least one supplier relationship and zero otherwise, to examine the extensive margin of separations. The results show that the strength of local clans significantly reduce the probability of relationship disruption in supply chains. In column 4, we replace the denominator in Eq. (1) for calculating separation rates by the number of total suppliers in the current year. The results remain unchanged.

Next, we turn to the compilation date of genealogies. *The General Catalog of Chinese Genealogies* only records the genealogies that have been stored so far, which may cause survival bias. To ensure the robustness of our results, we use the number of genealogies compiled in different time periods to measure clan density. In column 1 of Panel B, we confirm our findings using the number of all genealogies recorded in each prefecture. In columns 2, we focus on the genealogies compiled before 1912, as in the late Qing dynasty, China experienced severe wars that might affect the storage and the compilation of genealogies (Fan et al., 2023). The establishment of the People's Republic of China in 1949 is a milestone of significant societal restructuring, officially abolishing clans and nullifying the rules of clans (Greif and Tabellini, 2017). In column 3, we count the number of genealogies before 1949 as the clan measure. The results across all columns in Panel B of Table 3 suggest that strong clan strength minimizes the risk of supply chain disruption.

Then, we perform additional robustness checks by clustering standard errors at the prefecture level instead of at the firm level, as firms located in the same district may exhibit similar behavioral patterns, and this is particularly relevant given that our clan variable is

Table 4
Robustness: clustering standard errors at the prefecture level.

	Model 1: Firm-Year Fixed Effect Model	Model 2: Cox Proportional Hazards Model	Model 3: Customer-Supplier-Year Fixed Effect Model	
	Separation Rate	Duration	Disruption dummy	
	(1)	(2)	(3)	(4)
Clan	-0.0042***	0.0511**	-0.0381***	-0.0236**
	(0.002)	(0.023)	(0.009)	(0.012)
Controls	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	-	Yes	Yes
Province-Year FE	Yes	-	Yes	Yes
Firm FE	-	-	Yes	Yes
Supplier-Year FE	-	-	-	Yes
Customer Industry Strata	-	Yes	-	-
Supplier Strata	-	Yes	-	-
Year Strata	-	Yes	-	-
Observations	26,299	6,071	31,556	23,004
Adj R^2	0.041		0.053	0.252

Notes: This table reports the estimations clustering standard errors at the prefecture level. In column1, we take separate rates defined in Eq. (1) as the dependent variable and estimate Eq. (2) using a firm-year panel. In column 2, we adopt the Cox Proportional Hazards Model and estimate the effect of clan density on relationship duration using a cross-sectional dataset. In columns 3-4, we use the customer-supplier-year dataset and take a dummy indicating the relationship disruption as the dependent variable. Columns 1 control for province-year, and industry-year fixed effects; column 3 adds a firm fixed effect; column 4 additionally control for supplier-year fixed effects. *Clan* is calculated as the logarithm of one plus the cumulative number of genealogies before 1990 normalized by the 1990 population for each prefecture. The definitions of other variables can be found in Appendix Table A1. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table 5
Robustness: different subsamples.

	Exclude treaty port cities (1)	Exclude southeast provinces (2)	Exclude ethnic autonomous regions (3)	Exclude municipality and province capitals (4)	Exclude POEs (5)	Exclude family firms (6)
Clan	-0.0043**	-0.0051***	-0.0042***	-0.0040**	-0.0062**	-0.0054**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17242	13592	25,060	20,197	13,037	13,404
Adj R^2	0.042	0.051	0.042	0.044	0.066	0.067

Notes: In this table, we exclude observations that are potential subject to identification issues. The dependent variables are the separate rates defined in Eq. (1). In column 1, we remove firms from treaty port cities. In column 2, we follow Fan et al. (2023) to exclude firms from nine provincial judications in the southeast, that is, Jiangsu, Shanghai, Zhejiang, Guangdong, Guangxi, Fujian, Hainan, Anhui, and Jiangxi provinces. In column 3, we exclude firms from five ethnic autonomous regions: Guangxi, Tibet, Ningxia, Inner Mongolia, and Xinjiang. In column 4, we exclude firms operating in municipality and province capitals. In column 5, we exclude private-controlled enterprises (POEs). In column 6, we exclude all family firms. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

defined at the district level. This may lead to biased standard errors if not properly accounted for. The estimations with an alternative clustering level are reported in Table 4.[2] Our main results virtually unchanged.

Fig. 1 clearly shows that the geographical distribution of clans is not random. Cities along the coastal line and in the southeast explicitly stored more genealogies. Meanwhile, these cities have high economic growth and complex industrial structure, and local firms are expected to be more involved in supply chains and exposed to a variety of exogenous shocks. To address this issue, we first exclude firms from cities that were treaty ports from 1840 to 1910. The treaty ports rode on the first wave to establish connections with the west and have primarily developed manufacturing industries in the modern era. The penetration of western norms may also vary between treaty and non-treaty ports cities. Column 1 of Table 5 reports the results. Again, clans play an important role to curb supplier separations. Furthermore, we follow Fan et al. (2023) to exclude firms from nine provincial jurisdictions along the southeast coast, that is, namely, Jiangsu, Shanghai, Zhejiang, Guangdong, Guangxi, Fujian, Hainan, Anhui, and Jiangxi provinces, where genealogies are concentrated. The results in column 2 show that even in regions with faint traditional culture, the presence of clans is still helpful in stabilizing firms' relationship.

[2] Thank an anonymous reviewer for this suggestion. When clustering at the industry level, we still obtain consistent results.

Table 6
Robustness: time-varying clan variable and its influence.

	Exclude moving firms	Duration of local operation		CEO	
		Long	Short	Local	Non-Local
	(1)	(2)	(3)	(4)	(5)
Clan	-0.0033*	-0.0070**	0.0008	-0.0075**	-0.0020
	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	21,401	11,634	9,512	8,130	17,921
Adj R^2	0.042	0.017	0.090	0.093	0.025

Notes: In this table, we discuss the time-varying nature of clan variable and its influence. The dependent variable is the separate rate, calculated separately for each subsample listed in the column headers. In column 1, we exclude firms that relocated in our sample period. In columns 2 and 3, we categorize firms into long- and short-duration subsamples based on whether their years of local operation are above or below the median. In columns 4 and 5, we divide subsamples based on whether firms have a local CEO. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

In columns 3 and 4 of Table 5, we further exclude firms from five ethnic autonomous regions and those from four municipalities and provincial capitals, respectively (Zhang, 2020; Cheng et al., 2021). The former group involve minorities and their religions, and the latter group may receive special government support. Excluding different subsamples does not alter our findings. Moreover, private-controlled firms, particularly family firms, are more susceptible to the influence of clan culture. These firms have relatively concentrated ownership and more depend on trust-based relationships, aligning closely with the norms and values embedded in clan traditions. To ensure that our main findings are not driven by such firms, we further exclude private-controlled firms and family firms in columns 5 and 6, respectively. The results remain virtually unchanged.

The core clan variable in our baseline model is time-varying, as several firms relocate and are exposed to varying levels of clan culture across regions. However, business relocation itself may disrupt supply chain relationships and bias our estimates. We perform a sensitivity check by excluding firms changing registration address in our sample period and report the results in column 1 of Table 6. This action does not alter our findings. In addition, even within the same region, firms may differ in the extent to which they are influenced by local culture. To test this, we conduct subsample analyses by separately classifying firms based on the years of local operation and whether they have a local CEO. The results of columns 2 – 5 suggest that the stabilization role of clans is more pronounced when firms has been embedded in the region for a considerable period or they have local CEOs. This implies that the influence of culture is also contingent on firm-specific characteristics that shape the extent of cultural exposure.

4.3. Instrumental variable approach

The strength of supplier-customer relationship is influenced by numerous factors that cannot be exhaustively controlled. To formally address the endogeneity concerns, we employ instrumental variable approach. Specifically, we choose three instruments (IV) for clan density.

The first IV is rice-wheat ratio, defined as the ratio of rice suitability to wheat suitability indices in each prefecture. Talhelm et al. (2014) propose the rice theory and demonstrate that a history of rice farming fosters interdependence, whereas wheat farming promotes independence. The characteristics of wetland rice cultivation contribute to cooperative norms (Zhou et al., 2023): first, rice is a labor-intensive crop, requiring approximately twice the labor input of wheat for the same land area; second, rice requires extensive irrigation, dredging, planting, weeding, transplanting, and careful field leveling, which necessitates cooperation among neighbors and villagers for the construction and maintenance of irrigation infrastructures; third, it is easy to identify and penalize people who shirk their responsibilities or freeload in the process of rice cultivation. The intensive cooperation in daily life can enhance the cohesions of belonging community, especially the clan in agricultural society. Also, the fight for water resources is important to rice cultivation, and the strength of clans largely determine the resource allocation. We extract the data of rice and wheat suitability indices from the FAO and IISAS's global agro-ecological zones (FAO and IIASA, 2021) database.

The second IV is the agricultural potential yield. As mentioned, crop cultivation requires community cooperation, which yet may vary with the levels of land productivity. Litina (2016) points out that high land productivity weakens local social capital and undermines collective actions. Clan culture is also characterized by collectivism. Intensified cooperation on barren land can further strengthen the sense of belongings among clan members. We collect the data of agricultural potential yield from Liu et al. (2015).

The third IV is the number of chaste women in each prefecture. Clans, often synonymous with patrilineages, are centered on patriarchy and establish a hierarchical ethical framework (Zhang, 2020). The feudal ethics manifests in gender relations by emphasizing women's chastity and imposing strict behavioral constraints. Particularly, Confucianism proposes the rule of three obediences for women, that is, obedient to the father, husband, and son. The chaste women, who remain loyal after the death of their husbands, were highly appreciated by both the state and local clans, particularly during the Ming and Qing dynasties (Theiss, 2005). Clans typically allocated resources to guarantee the living conditions of chaste women, and sometimes forced widows to remain chaste for protecting clan reputation. In this way, the number of chaste women reflects the intensity of clan culture in a region. We collect the

Table 7
Instrumental variable estimations.

	First stage Clan (1)	**Second stage** Separation Rate (2)
Clan		-0.0152**
		(0.007)
Rice Wheat Ratio	0.0171**	
	(0.008)	
Agricultural Potential Yield	-0.3640***	
	(0.087)	
Chaste Women	0.1636***	
	(0.037)	
Controls	Yes	Yes
Industry-Year FE	Yes	Yes
Province-Year FE	Yes	Yes
Kleibergen-Paap rk LM statistic	57.46	
Kleibergen-Paap Wald rk F statistic	22.20	
Hansen test P-value	0.6802	
Centered R^2		0.006
Observations	26,283	26,283

Notes: This table presents two-stage-least-squared estimations for the relation between the strength of local clans and separation rates. We use three instruments. The first one is the ratio of rice to wheat suitability indices, the data of which are collected from the Global Agro-Ecological Zones (FAO and IIASA, 2021) database. The second one is the logarithm of Chinese agricultural potential yield from Resource and Environment Science and Data Center of Institute of Geographic Science and Natural Resources Research, Chinese Academy of Sciences. The third one is the logarithm of one plus cumulative number of Chaste Women during Ming and Qing dynasties from the Official History of the Ming Dynasty and the Draft History of Qing (Dong and Zhang, 2024). The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

name and location of each chaste woman from the *Official History of the Ming Dynasty* (in Chinese, *ming shi*) and the *Draft History of Qing* (in Chinese, *qing shi gao*).[3]

To formally test the relevance condition, we regress each prefecture's density of genealogies on the three IVs and report the results in column 1 of Table 7. Consistent with our expectations, clan density is positively correlated with rice-wheat ratio but negatively correlated with land productivity. Stronger clans are also associated with more chaste women. The IV estimations are reported in column 2. We find that the strength of local clans has a strong explanatory power for the resilience of supply chains. The IV estimated coefficient on the instrumented clan variable is three times the OLS estimates (0.0152 vis-à-vis 0.0042), implying that our endogenous explanatory variable may underestimate the impact of clan strength. The F-statistics and Hansen J statistics reveal that our IVs pass both weak instrument and over-identification tests. Overall, the IV estimations reinforce our baseline findings that clans help firms stabilize their relationship with suppliers.

5. Clan and supply chain resilience under exogenous shocks

So far, we have provided preliminary evidence that firms in clan-intensive regions maintain sticky supply chain relationship. However, we still lack evidence on whether and how clans can make supply chains more resilient to exogenous shocks. A myriad of shocks can cause supply chain disruptions, which lead to input shortage and production impediment. For example, the Great East Japan Earthquake of 2011 attracts much attention from scholars, pointing out that the disruption caused by earthquake can be transmitted into sizable fluctuations at the macroeconomic level (Carvalho et al., 2021). Considering the substantial costs of supply chain disruptions for both firms and regions, it is important to investigate the resilience value of clans in the face of various shocks. In this section, we choose three types of shocks that have been proven to increase the vulnerability of supply chains in existing literature: natural disaster, tariff, and credit shocks.

5.1. Natural disaster shock: earthquake

Earthquake is typically considered as an exogenous shock to disrupt input-output linkages (Boehm et al., 2019). It not only affects production ability but also financial and stock performance. Supply chain partners are likely to terminate relationship with firms in disaster-stricken areas, because, on the one hand, business exchange cannot be guaranteed on schedule, and on the other hand, firms in

[3] Special thanks to Yu Zhang for sharing the data of chaste women. The detailed procedures for data collection can be found in Dong and Zhang (2024).

Table 8
Clans and supply chain resilience: earthquake shocks.

	Separation Rate	
	(1)	(2)
Clan × Earthquake		-0.0049***
		(0.002)
Earthquake	0.0020*	0.0023
	(0.001)	(0.002)
Controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
City FE	Yes	Yes
Observations	22,616	21,765
Adj R^2	0.037	0.038

Notes: The dependent variable in this table is the separation rate defined in Eq. (1). *Earthquake* is an interaction term combing a treatment variable indicating that a firm's locality is hit by any earthquake and zero otherwise, and a post dummy indicating five years following the earthquake. The estimations draw on a sample consisting of rolling event windows of [-5, 5] where year 0 is the earthquake year. The control variables are the same as those in Table 2. All regressions include city, industry, and year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

affected areas may face liquidity shortage due to factory and equipment repairs, suspension of production, or additional subsidies to employees. Clans are expected to support firms in such difficult periods. The strength of social capital facilitates collective actions against losses from natural disasters and expedites the quick restoration of local economic and social activities. Firms may obtain informal financial resources, for example, short-term private credit, to overcome liquidity shortage.

To examine the beneficial role of clans in the periods of natural disasters, we collect the data on earthquakes, including the date, magnitude, and location, from the CSMAR database. As illustrated in Section 3.2, we employ a difference-in-differences (DD) framework to estimate the effect of earthquakes on supply chain disruptions and the moderating effect of clans. For each earthquake, we build an event window of [-5, 5], with year 0 being the earthquake year. We also classify firms into the treatment and control groups according to whether they operate in disaster-stricken areas, and particularly we require that control firms are not affected by any earthquakes throughout the entire sample period. Then, we have a DD term of $Earthquake_{ict}$ capturing the interaction between treatment status and post-event period.

Table 8 reports the results. In column 1, we first confirm whether earthquake shocks indeed disturb supply chains of local firms. The results show that firms in disaster-stricken areas experience significantly higher separation rates, clarifying natural disasters as a source of supply chain instability. Then, we incorporate the interaction term between the clan variable and the DD term of $Earthquake_{it}$ in column 2. The results show that the estimated coefficient before the interaction is negative and statistically significant at the level of 1%, suggesting that clans can reduce the disaster-caused disruptions. The setting of earthquake underscores that clans make local firms' supply chains more resilient to natural disaster shocks.

5.2. *Credit shock: the guideline on green credit*

We turn our attention to credit policy shocks in this subsection. Trade credit is a focal point in supply chain literature. Financial constraints can incentivize firms to participate in supply chains, as trading partners allow for delayed payment of products and provide liquidity even when banks are not willing to extend loan resources (Minetti et al., 2019). Trade credit functions as a glue to secure a supply chain relationship. Ersahin et al. (2024) show that firms that are affected by natural disasters, including blizzards, earthquakes, floods, and hurricanes, receive increased trade credit from suppliers; notably, partners choose to terminate relationships with affected firms if financial constraints inhibit the extension of trade credit. In this way, we posit that liquidity shortage affects the stability of firms' supply chains.

We exploit the *Guideline on Green Credit* (henceforth *Guideline*) issued by the former China Banking Regulatory Commission (CBRC) in 2012, as an exogenous credit shock to the liquidity of targeted firms. According to the *Guideline*, firms operating in heavily polluting, resource-intensive, and over-capacity sectors, referred to as "two high, one overcapacity", face restrictions in accessing bank credit. Firms in targeted industries received fewer bank loans and paid higher interest rates (Fan et al., 2021), which both lead to tightened credit constraints. Bank credit shocks can transmit along supply chains and affect the performance of downstream and upstream firms, notably reducing the trade credit (Alfaro et al., 2021; Costello, 2020). This may motivate partners to sever relationship with affected

Table 9
Clans and supply chain resilience: credit shocks.

	Separation Rate	
	(1)	(2)
Clan × GCguide		-0.0271***
		(0.008)
GCguide	0.0337**	-0.0297
	(0.016)	(0.025)
Controls	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
City FE	Yes	Yes
Observations	16,109	15,541
Adj R^2	0.051	0.050

Notes: This table examines the impact of credit policy shocks and whether clans enhance resilience to the credit shock, drawing on an estimation window of [-5, 5]. *GCguide* is an interaction term combining a treatment dummy which equals one if the firm is operating in industries targeted by the *Guideline on Green Credit* issued by CBRC in 2012 and 0 otherwise, and a post dummy for the five-year post-policy period. The specific industries are listed in footnote 3. The control variables are the same as those in Table 2. All regressions include city, industry, and year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table 10
Clans and supply chain resilience: tariff shocks.

	Domestic Separation Rate		Foreign Separation Rate	
	(1)	(2)	(3)	(4)
Clan × Trade Shock		-0.0175*		-0.0094
		(0.009)		(0.021)
Trade Shock	-0.0303**	-0.0643***	0.0531*	0.0393
	(0.015)	(0.017)	(0.030)	(0.043)
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Observations	18,438	17,767	18,438	17,767
Adj R^2	0.022	0.023	0.015	0.015

Notes: The dependent variables in Columns 1-2 and columns 3-4 are separation rates of domestic suppliers and separation rates of foreign suppliers, respectively. *Trade Shock* is an interaction term combining a treatment dummy which equals one if the firm has at least one U.S. supplier whose products are targeted in China's retaliation tariff list during 2018-2019 and zero otherwise, and a post dummy indicating post-trade-war periods. The control variables are the same as those in Table 2. All regressions include city, industry, and year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

firms.

Likewise, we adopt the DD approach and classify firms in industries targeted by the *Guideline* as the treatment group[4] and others as the control group, and use an estimation window of [-5, 5], with year 0 representing the policy implementation year. $Shock_{it}$ in Eq. (3) now refers to $GCguide_{it}$, a DD term combining a treatment dummy for the green credit policy and a post dummy for years after the implementation. Column 1 of Table 9 reports the estimates on how the credit policy affects the supply chains of targeted firms. As expected, we find that firms affected by the Guideline's restrictions on bank credit exhibit significantly higher separation rates.

Clans have been shown to provide informal financing channels and support the development of private sector (Chen et al., 2022; Zhang, 2020). We conjecture that clans can facilitate financing for constrained firms through their network and thus alleviate the adverse impact of formal credit policies. The estimations of Eq. (3) are reported in column 2. The interaction term of clan variable and credit shock is statistically and significantly negative, implying that strong clan strength enhances supply chain resilience to credit shocks by reducing relationship disruptions.

[4] Following the methodology in Lei et al. (2023), we categorize firms belonging to Class A environmental and social risk industries listed in "*Key Evaluation Indicators of Green Credit Implementation*" as targeted firms. The industries include nuclear plants, water conservancy and inland port engineering and construction, hydroelectric generation, natural gas mining, coal mining and washing, nonmetallic mining and dressing, nonferrous metal mining and dressing, ferrous metal mining and dressing, and other mining industries.

5.3. Trade Shock: tariff increases

Trade shocks are considered one of the major sources of supply chain disruptions (Grossman et al., 2023). The unexpected outbreak of the China-U.S. trade war signals an unprecedented era of heightened trade policy uncertainty and a reconfiguration of global supply chains. Starting in July 2018, the U.S. government greatly increased tariffs on imports from China, targeting $50 billion, $200 billion, and $300 billion worth of goods in consecutive three rounds. As a response, the Chinese government implemented retaliation tariffs on goods imported from the U.S., which approximately increased the average tariff on the U.S. goods from 7.8% in 2017 to 25.9% in 2019. The great tariff hikes undoubtedly increase production costs and promote firms to renegotiate terms with existing suppliers or turn to new suppliers. As the focus of this paper is firms' supplier disruptions, that is, sourcing decisions, here we exploit the implementation of Chinese retaliation tariffs as an exogenous shock and examine its impact on supply chains.

For this tariff shock, we also employ a difference-in-differences setting in which the treatment group consists of firms with at least one U.S. supplier whose products are included in China's retaliation tariff list, while the control group comprises other firms. The DD term in tariff shock regressions, $tradeshock_{it}$, is a combination of a treatment dummy indicating the exposure to input tariff increases and a post dummy for years after tariff increases. Unlike previous settings, here we further categorize suppliers into foreign and domestic ones to observe the restructuring of supply chains in the face of a cross-border shock.

In columns 1 and 3 of Table 10, we separately examine the impact of exposure to input tariff increases on supply chain disruptions. The results show that firms choose to terminate more foreign suppliers but maintain relatively stable domestic supply chains as a response. This implies that firms substitute between foreign and domestic suppliers to maintain production in response to tariff shocks, which depicts a negative impact on supply chain stability.

Next, we estimate Eq. (3) and pay particular attention to the interaction between clan density and the trade shock variable. In columns 2 and 4, we find that clans significantly reduce domestic separation rates for tariff-affected firms, but we find no evidence on foreign separation rates. The contrast is reasonable. The informal markets for financing or social resources provided by clan network are largely ineffective in foreign countries. This suggests that clans can stabilize domestic supply chains or facilitate re-shoring to support firms in the face of shocks arising from international business.

Taken together, our investigation demonstrates that strong clan strength increases the stickiness of supply chain relationships for local firms and particularly, enhances supply chain resilience to various types of exogenous shocks.

6. Mechanisms

We explore two potential mechanisms through which clans reduce supply chain disruptions in this section. First, clans improve the contractual environment and increase the reliability of local firms in business relationships. Second, social capital embedded in clans enables firms to stabilize socially connected supply chain relationships. The two mechanisms are closely related to characteristics of clans, including the formation of social capital, collectivism actions, and moral obligations that discourage opportunistic behaviors.

6.1. Contractual environment

Institutional environment is clearly an important factor in establishing business relations (Contractor et al., 2020; Slesman et al., 2021). According to the contract theory proposed by North (1981), institutions provide legal and social frameworks that facilitate the enforcement of private contracts. A supply chain relationship involves a supplier and a customer with productive exchange, which necessitates substantial upfront and customization costs for the maintenance. However, incomplete contracts often lead to frequent renegotiation and hinder deep cooperation, thereby increasing transaction costs and disrupting the stability of a relationship (Grossman et al., 2024). The problem might be more prominent in China where legal institutions are underdeveloped. According to the World Justice Project 2023, China ranks 97th across 142 countries in the world on the Rule of Law Index.[5] Opportunistic behaviors undermine the formation of durable business relationships.

Clans are organizations that enforce informal rules, sustain cooperation, and exhibit a strong sense of collectivism (Fan et al., 2023). We conjecture that clans can act as effective informal arrangements to improve contracting institutions. The code of conduct explicitly outlined in the genealogies monitors the behaviors of clan members and prevents them from harming outsiders in a way that damage the clan's reputation (Liu, 1959). This reduces the engagement in opportunistic behaviors of people from regions with strong clan culture. A relationship involves two parties, and either party can initiate the termination. From another perspective, collectivism hinders local firms' counterparts from breaking contracts and jeopardizing the relationship, as this action might be considered as damaging the interests of the whole clan and create obstacles for cooperation with other local firms in the future.

To test our hypothesis, we extend the baseline Eq. (2) by incorporating a proxy for local contracting institutions and its interaction with the clan variable. Specifically, we count the number of legal cases in each prefecture, and roughly assume that fewer cases represent more favorable contractual environment, as business disputes are less likely to arise. By collecting the textual data on legal case involving listed firms from the CSMAR database, we manually identify whether each case is related to contractual disputes, to ensure the relevance of our measure to the contractual environment. For example, intellectual property theft is considered contract-related, whereas the dismissal disputes are not. We also adopt a broader measure for institutional environment by counting the number

[5] See details on the website of https://worldjusticeproject.org/rule-of-law-index/country/2023/China/.

Table 11
Mechanism: contractual environment.

	Log # contract dispute cases (1)	Log # arbitration and litigation cases (2)
Clan	-0.0012	-0.0002
	(0.002)	(0.002)
Legal Cases	-0.0017	-0.0008
	(0.002)	(0.002)
Clan × Legal Cases	-0.0021***	-0.0017***
	(0.001)	(0.001)
Controls	Yes	Yes
Industry-Year FE	Yes	Yes
Province-Year FE	Yes	Yes
Observations	26,299	26,299
Adj R^2	0.042	0.042

Notes: The dependent variable in this table is the separation rate defined in Eq. (1). *Case* in columns 1 and 2 are the logarithm of the number of contractual dispute cases and the number of arbitration and litigation cases in each city, respectively. The data on the legal cases are collected from the CSMAR database. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table 12
Mechanism: social network.

Panel A: Social network			Baseline Sample			
			Connected		Non-connected	
			(1)		(2)	
Clan			-0.0046***		0.0004	
			(0.002)		(0.000)	
Controls			Yes		Yes	
Industry-Year FE			Yes		Yes	
Province-Year FE			Yes		Yes	
Observations			25,264		25,264	
Adj R2			0.038		0.029	
Panel B: Social network under exogenous shocks						
	Earthquake		Credit Shock		Tariff Shock	
	Connected	Non-connected	Connected	Non-connected	Connected	Non-connected
	(1)	(2)	(3)	(4)	(5)	(6)
Clan × Shock	-0.0048***	-0.0004	-0.0260***	0.0009	-0.0149	-0.0014
	(0.002)	(0.000)	(0.008)	(0.005)	(0.011)	(0.003)
Shock	0.0025	0.0002	-0.0290	0.0047	-0.0594***	-0.0059
	(0.002)	(0.001)	(0.022)	(0.016)	(0.021)	(0.007)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,876	20,876	15,173	15,173	16,849	16,849
Adj R2	0.035	0.005	0.044	0.004	0.018	0.003

Notes: In Panel A, the dependent variables in Columns 1and 2 are the separation rates of connected suppliers and non-connected suppliers, respectively. We classify a supply chain relationship as connected if the CEO or chairpersons of the customer and supplier share hometown ties. In Panel B, we examine the differential effects of clans in helping firms withstand three types of shocks between connected suppliers and non-connected suppliers. *Shock* refers to the DD terms defined in the analyses of earthquake, credit policy, and trade shocks. The estimations are conducted using a window of [-5, 5], where year 0 is the year of shock. The control variables are the same as those in Table 2. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

of arbitration and litigation cases in each prefecture.

The estimation results drawing on the firm-year panel are reported in Table 11. Both the formal and informal institutions reduce the supplier disruptions, though the coefficients are insignificant. More importantly, in both columns, the estimated coefficients on the interaction term are significantly negative, suggesting that clans play a more important role in stabilizing supply chains for firms in regions with poor institutions. The pattern supports the statement in Acemoglu and Johnson (2005) that entities can mitigate the adverse effect of bad legal frameworks by developing alternative informal arrangements. Overall, the evidence shows that clans can serve as a substitute for contractual institutions.

6.2. Social network

Social connections are crucial to the establishment and stability of supply chain relationship (Chen et al; 2021; Dasgupta et al., 2021; Ding et al., 2023). Through the information-sharing function and the "network reputation" effect, interpersonal connections among executives of supply chain partners can mitigate the issues associated with incomplete contracts, alleviate hold-up problems, and facilitate deep cooperation, all of which contribute to durable relationships. Studies often examine social connections through the lens of studying and working experience in developed countries (Fracassi and Tate, 2012). In China, clans can form the foundation for social capital (Cao et al., 2022). The bond to a clan is inherently determined and endures throughout one's lifetime. In such as family- and relationship-centered society, the social capital embedded in clans provides effective channels for seeking business partners. Also, individuals who grow up in regions with strong clan culture are more likely to value relationships and make efforts to maintain them. These factors facilitate mutual trust in business exchange.

To examine whether clans stabilize supply chains through social networks, we first categorize each firm's suppliers into connected and non-connected based on whether the CEO or chairman of two partners share hometown ties. Then, we separately calculate the separation rates of connected and non-connected suppliers and estimate Eq. (2). The results are reported in columns 1 and 2 of Panel A, Table 12, respectively. We find that the stabilization effect of clans only holds for socially connected suppliers, while the strength of clans has a loose relationship with the separation rate of non-connected suppliers.[6] This contrast underscores the importance of social connections in the influence of clans. Individuals from the same regions with intensive clan culture share common values and beliefs and adhere to similar codes of conduct, which strengthens the resilience of the relationships among their managing firms.

In addition, personal names are key indicators of one's identity (Pilcher, 2016), and surname sharing can be a form of connectedness that foster a sense of in-group identification. Thus, we categorize each firm's suppliers into surname-sharing and non-surname-sharing groups based on whether the CEO or chairman of two partners share the same surnames. The results in Appendix A3 suggest that the stabilization effect of clans holds only within the surname-sharing subsample.[7] Taking a step further, we combine hometown and surname connections and divide the sample into four subsamples: hometown- and surname-sharing, hometown-sharing and non-surname-sharing, non-hometown-sharing and surname-sharing, and non-hometown-sharing and non-surname-sharing, and estimate the separation rate regressions in each subsample. Unsurprisingly, the clan is most effective at reducing supply chain disruptions among suppliers with both hometown and connections.

To further validate the social network mechanism, we revisit three types of shocks discussed in Section 5.1 and examine whether the capacity of clans to withstand external shocks differs between socially connected and non-connected subsamples in Panel B of Table 12. By taking the separation rates for connected and non-connected suppliers as dependent variables— columns 1 and 2 for earthquakes, columns 3 and 4 for the credit policy shock, and columns 5 and 6 for the tariff shock—we find that clans only mitigate disruptions caused by earthquake and credit policy shocks within connected supply chains, whereas imposing negligible impact on the stability of non-connected ones. However, we do not observe a clear differential effect in the case of tariff shocks. Overall, these results highlight the critical role of social capital in underpinning the resilience value of clans within supply chains, particularly when firms are exposed to exogenous shocks in domestic markets.

7. Further discussion

So far, we have provided robust evidence that clans help firms stabilize their supply chains, particularly in the face of exogenous shocks. In this section, we address two additional questions that further clarify the role of clans in supply chain management. The first is whether clans can enhance firms' capacity of post-disruption recovery. While the central focus of our analysis has been on how to prevent disruptions, that is, ex-ante stabilization, it is also important to investigate the role of clans in ex-post remediation. The other is to discuss the real effect of supply chain disruptions. The fundamental assumption underlying all the analyses is that frequent supply chain disruptions are harmful to firm operations and thereby, clans provide crucial resilience value. However, studies based on the U.S. data (e.g., Xu et al., 2025) have shown that great supplier churn may enhance firm performance, such as boosting sales. Thus, we aim to explore the relationship between supply chain instability and firm performance in the context of the Chinese market.

[6] To mitigate the concern that clan culture itself may influence the number of linked supply chain relationships, we employ the number of terminated suppliers, instead of the separation rate, to measure supply chain instability, and the results remain unchanged for the social network analysis. Furthermore, when using the customer-supplier-year dataset and take the disruption dummy as the dependent variable, we still find that hometown ties can strengthen the stabilization effect of clan culture. For brevity, we do not tabulate results in the main text which could be provided upon request.

[7] Psychological literature documents that the role of surnames in network formation depends on their rarity (Jacobs, 1979). Thus, individuals sharing a rare surname tend to exhibit a stronger sense of affinity. To further explore the role of surnames, as well as the clans' stabilization effect, we perform heterogeneity analysis and classify a CEO's surname as large surnames based on two criteria: i) the surname ranks among the top three by the number of clan genealogies in the firm's locality; ii) the surname ranks among the top three most frequent among supplier CEOs. To save space, we report the results in Appendix Table A4. Clans are more important when CEOs belong to small surnames, consistent with a "small-group affinity" phenomenon in Even-Tov et al. (2023) and Tan et al. (2021).

Table 13
Clans and post-disruption recovery.

Panel A: Number			
	(1) New supplier	(2) Terminated supplier	(3) Total
Clan	-0.0241**	-0.0219**	-0.0405
	(0.012)	(0.010)	(0.027)
Controls	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes
Observations	6,470	6,470	6,470
Adj R^2	0.198	0.135	0.165
Panel B: Rate			
	(1) Entry	(2) Termination	(3) Churning
Clan	-0.0113*	-0.0147**	-0.0101***
	(0.007)	(0.007)	(0.004)
Controls	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes
Observations	6,470	6,470	6,470
Adj R^2	0.053	0.106	0.077

Notes: This table presents estimation results of the supply chain recovery dynamics following a supply chain disruption. We restrict the sample to firms that experienced supply chain disruptions and limit the analysis to periods following those events. In columns 1-3 of Panel A, the dependent variables are the logarithm of the number of new suppliers, the number of terminated suppliers, and the number of total suppliers, respectively; in columns 1-3 of Panel B, the dependent variables are the entry rate, termination rate, and churning rate that is defined as the minimum value of entry and termination rates. All regressions include industry-year and province-year fixed effects. The control variables are the same as those in Table 2. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table 14
Supply chain instability and firm performance.

	Profit Margin			
	(1)	(2)	(3)	(4)
Separation Rate	-0.0071**			-0.0076**
	(0.003)			(0.004)
Entry Rate		0.0024		0.0030
		(0.004)		(0.004)
Churning Rate			-0.0087	
			(0.005)	
Controls	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
Observations	26,284	26,284	26,284	26,284
Adj R^2	0.583	0.583	0.583	0.583

Notes: The dependent variable is profit margin. *Separation rate* is the number of terminated suppliers scaled by the total number of suppliers; *Entry rate* is the number of new suppliers scaled by the total number of suppliers. *Churning rate* is the minimum of separation and entry rates. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

7.1. Post-disruption recovery

Supply chain resilience not only reflects a firm's ability to withstand shocks and reduces the likelihood of disruption, but also its capacity for rapid recovery and adaptability. To further examine the role of clan culture during post-disruption periods, we restrict the sample to firms that experienced supply chain disruptions and limit the analysis to periods following those events, and thereby, examine whether clan culture can facilitate the search for new suppliers after a disruption, and enhance firms' ability to return to its original state.

We construct two sets of variables to measure the dynamic recovery after disruptions: the first set includes the number of new suppliers, the number of terminated suppliers, and the total number of suppliers (in logarithm terms); the second set includes the rates of entry and termination, as well as the churning rate defined as the minimum value of entry rate and termination rate which captures the change in the portfolio of suppliers (Xu et al., 2025). Taking the first set as the dependent variables, we estimate Eq. (2) and report the results in Panel A of Table 13. We find that strong local clans cannot help firms effectively establish new relationships after supply chain disruptions, as the estimated coefficient on the clan variable in the regression of new suppliers is significantly negative and that in the regression of total suppliers is insignificant. Panel B provides consistent patterns: after disruptions, firms experience significantly lower entry rate and churning rate in their supply chains. These findings suggest that the influence of clans lies primarily in stabilizing existing supplier relationships, rather than in enhancing firms' ability of ex-post remediation.

7.2. Supply chain churning and firm performance

In this subsection, we examine the influence of supply chain adjustment on firm operations. We primarily focus on profit margin as a comprehensive proxy for operating performance, and estimate the specification as follows:

$$profitmagin_{i,t} = \beta_0 + \beta_1 x_{i,t} + Controls_{i,t-1} + \gamma_{jt} + \delta_{pt} + \varepsilon_{it}, \tag{4}$$

where

$$x_{i,t} \in \{Separation\ Rate_{i,t}, Entry\ Rate_{i,t},\ Churning\ Rate_{i,t}\},$$

Separation rate is defined in the same way as in the baseline model. Entry rate is the number of new suppliers scaled by the total number of suppliers. Churning rate is the minimum of separation and entry rates (Xu et al., 2025).

The estimation results are reported in Table 14. In columns 1-3, we sequentially include separation rate, entry rate, and churning rate in the regressions. We find that an increase in separation rates is associated with lower profit margin, but entry rates and churning rates have negligible impacts. In column 4, we simultaneously incorporate separation rates and entry rates into estimations, and again, we find that separation rates have a significantly negative impact on firm profitability. Moreover, in unreported results, when using alternative proxies for supply chain adjustment (the number of terminated suppliers, the number of new suppliers, or the change in the total number of suppliers) and alternative measures of firm performance (cost positions) consistently shows that greater supply chain instability is associated with worse performance.

These results imply that, in the China's context, higher separation rates within supply chains are indeed detrimental to firm operations and the resilience value created by clan culture are crucial.

8. Conclusions

This paper examines the role of clan culture in stabilizing supply chains of local firms. Specifically, we show that firms in regions with strong clan strength experience significantly fewer supplier disruptions. We corroborate this finding using the cross-sectional dataset and supplier-customer-year dataset and show that clan density can lengthen the duration of relationship and reduce the probability of disruption. After controlling for a variety of firm and city controls, performing a series of robustness checks, and adopting an instrumental variable approach, we interpret our results as a causal effect of clan culture on supply chain stability.

To further assess the resilience value provided by clans, we examine how firms adjust their supply chains in response to various shocks across regions with different levels of clan density. We show that earthquake, credit policy, and tariff shocks all negatively affect the stability of supply chains; notably, strong clans can enhance supply chain resilience to these shocks.

We explore two mechanisms: contractual environment and social network. We show that clans play a more important role in stabilizing supply chains in regions with weak institutional environment. The pattern echoes Acemoglu and Johnson (2005)'s insight that individuals can develop alternative informal arrangement to mitigate the adverse effect of poor contracting institutions. We also discuss the role of social connections in stabilizing supply chains. In both ordinary and challenging periods, clans primarily enhance the resilience of socially connected supply chains.

Our work contributes to the burgeoning literature on supply chains from the perspective of social norms and culture. Secure and resilient supply chains are essential to the prosperity of firms and the overall economy. While studies primarily focus on the factors that motivate the establishment of supply chain relationships and disrupt supply chains, we offer insight into how to enhance the resilience of existing links. This is particularly critical during periods of heightened geopolitical risks and a multitude of disruption causes (Grossman et al., 2023). We also complement the literature on the persistent effect of historical institutions and cultural traits. Firms in China and other emerging countries are rapidly integrating into supply chains. While the clan is a typical Chinese informal organization, our findings on the improvement of contracting institutions and the formation of social capital in stabilizing supply chain can be extended to other contexts.

Declaration of competing interest

The authors have nothing to disclose.

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Appendix

Appendix Table A1, A2, A3, A4

Table A1
Definitions of main variables and Data source.

Variables	Definitions	Data source
Dependent variables		
Separation Rate	The number of terminated suppliers divided by the average number of suppliers in the current and preceding years.	Factset Revere
Duration	The number of years that a given supplier-customer relationship is maintained.	Factset Revere
Disrupt	A dummy indicating whether a given supplier-customer relationship is terminated in a specific year.	Factset Revere
Independent variables		
Clan	The logarithm of one plus the cumulative number of genealogies compiled before 1990 normalized by the 1990 population in the firm's locality prefecture.	The General Catalog of Chinese Genealogies
Control variables		
Age	Firm age	CSMAR
Foreign	A dummy indicating whether the firm has foreign ownership.	CSMAR
SOE	A dummy indicating whether the firm has state ownership.	CSMAR
Dual	A dummy indicating a CEO-chairman duality	CSMAR
Leverage	Total liabilities divided by total assets.	CSMAR
ROE	Net profits divided by equity.	CSMAR
Profit	Operating profits divided by revenue.	CSMAR
Size	The logarithm of total number of employees.	CSMAR
Cash	Cash divided by total assets.	CSMAR
Fixed Asset	Net fixed assets divided by total assets.	CSMAR
First Holder	The percentage of shares held by the largest shareholder.	CSMAR
Land Size	The logarithm of the area.	Resource and Environment Science and Data Center
ln Distance to Coast	The logarithm of the nearest distance from prefecture *c*'s centroid to the coast.	From Chen et al. (2020)
River Density	The length of river within the jurisdiction of prefecture *c* scaled by area.	National Geomatics Center of China
Slope	The average slope of prefecture *c*.	Resource and Environment Science and Data Center
ln GDPpc	The logarithm of GDP per capita.	China City Statistical Yearbook
Loan/GDP	Balance of deposits in financial institutions divided by GDP.	China City Statistical Yearbook
ln General Budget Expenditure	The logarithm of one plus the amount of fiscal expenditure	China City Statistical Yearbook
ln Education Expenditure	The logarithm of one plus local finance general budget expenditure on education.	China City Statistical Yearbook
ln Firm Number	The logarithm of one plus the number of industrial enterprises above scale	China City Statistical Yearbook
Instrument variables		
Agricultural Potential Yield	The logarithm of agricultural potential yield in each prefecture.	Resource and Environment Science and Data Center
Chaste Women	The logarithm of one plus cumulative number of chaste women during Ming and Qing dynasties in prefecture *c*.	From Dong and Zhang, (2024).
Rice Wheat Ratio	$Ratio_i = ln\left(10000 \times \frac{Wetland\ Rice\ Suitability_{i,c} + 0.01}{Wheat\ Suitability_{i,c} + 0.01}\right).$	Global Agro-Ecological Zones (FAO and IIASA, 2021) database
Other variables		
Clan all	The logarithm of one plus the number of all the genealogies normalized by population of 1990 of prefecture *c*.	The General Catalog of Chinese Genealogies
Clan before 1912	The logarithm of one plus the cumulative number of genealogies written before 1912 normalized by population of 1990 of prefecture *c*.	The General Catalog of Chinese Genealogies
Clan before 1949	The logarithm of one plus the cumulative number of genealogies written before 1949 normalized by population of 1990 of prefecture *c*.	The General Catalog of Chinese Genealogies
Earthquake	An interaction term combing a treatment variable indicating the magnitude of earthquake in a firm's location and a post dummy for five years following the earthquake.	CSMAR
Trade Shock	An interaction term between a treatment dummy indicating whether a firm has at least one U.S. suppliers whose products are included in China's retaliation tariff list during 2018-2019 and a post dummy for years after tariff increases.	Factset Revere; Bown et al. (2021)
GCguide	An interaction term between a treatment dummy indicating whether a firm is operated in industries targeted by the *Guideline on Green Credit* issued by CBRC in 2012 and 0 otherwise, and a post dummy for five-year post-policy period.	"*Key Evaluation Indicators of Green Credit Implementation*" by CBRC
Legal Cases	The logarithm of one plus the cumulated number of contractual dispute cases or total arbitration and litigation cases involving listed companies in a given prefecture.	CSMAR

(*continued on next page*)

Table A1 (*continued*)

Variables	Definitions	Data source
Entry Rate	The number of new suppliers divided by the average number of suppliers in the current and preceding years.	Factset Revere
Churning Rate	The minimum value of *Separation Rate* and *Entry Rate*.	Factset Revere

Table A2
Clan strength and new entries.

	Entry Rate	Log Number	IHS	Dummy	Entry Rate (Normalized by the number of existing suppliers)
	(1)	(2)	(3)	(4)	(5)
Clan	-0.0056*	-0.0083**	-0.0107**	-0.0092***	-0.0083**
	(0.003)	(0.004)	(0.005)	(0.003)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes
Observations	26,299	26,299	26,299	26,299	26,299
Adj R^2	0.058	0.142	0.142	0.134	0.082

Notes: The dependent variable in column 1 is *Entry Rate*, which is calculated as the number of new suppliers divided by the average number of suppliers in the current and preceding years. The dependent variables in columns 2-3 are the natural logarithm form and HIS form of the number of new suppliers. The dependent variable in column 3 is an indicator of whether the firm has established a new supplier. The dependent variable in column 5 is again the entry rate but calculated as the number of new entries divided by the number of suppliers in the current year. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table A3
Clan, surname connections, and separation rate.

	Separation Rate					
	Surname sharing	Non surname sharing	Hometown and surname sharing	Hometown- and non-surname-sharing	Non-hometown- and surname-sharing	Non-hometown- and non-surname sharing
	(1)	(2)	(3)	(4)	(5)	(6)
Clan	-0.0043***	0.0000	-0.0047***	0.0001	0.0004	-0.0000
	(0.002)	(0.000)	(0.002)	(0.000)	(0.000)	(0.000)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	26,299	26,299	25,264	25,264	25,264	25,264
Adj R2	0.038	-0.019	0.034	-0.020	0.028	-0.025

Notes: The dependent variable is the separation rate, calculated separately for each subsample listed in the column headers. In columns 1 and 2, we categorize each firm's suppliers into surname-sharing and non-surname-sharing based on whether the CEO or chairman of two partners have the same surnames. In columns 3 – 6, we split the full sample into four subsamples based on hometown and surname connections. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Table A4
Clan, rare surnames, and separation rate.

	Separation rate			
	Top 3 surname of the firm locality		Top 3 surname among suppliers	
	Large	Small	Large	Small
	(1)	(2)	(3)	(4)
Clan	-0.0025	-0.0039**	-0.0037	-0.0036**
	(0.005)	(0.002)	(0.005)	(0.002)
Controls	Yes	Yes	Yes	Yes
Industry-Year FE	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes
Observations	3,357	22,661	3,463	22,566
Adj R^2	-0.002	0.042	0.047	0.040

Notes: The dependent variable is the separation rate, calculated separately for each subsample listed in the column headers. In columns 1 and 2, we classify a firm' CEO as belonging to the large surname group if the surname ranks among the top three by the number of clan genealogies in the firm's locality and others as belonging to the small surname group. In columns 3 and 4, we divide samples based on whether the surname ranks among the top three most frequent among supplier CEOs in our sample. The control variables are the same as those in Table 2. All regressions include industry-year and province-year fixed effects. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance levels at 1%, 5%, and 10%, respectively.

Data availability

Data will be made available on request.

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